

Improving Stock Price Prediction with Enhanced Sentiment Integrated with Generative Adversarial Network

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Abstract

An advanced approach for stock price prediction using GANs along with very sophisticated techniques for sentiment analysis to make prediction accuracy even better is ESI-GAN. This assimilates a broad variety of up-to-the-minute sentiment and financial information extracted from news articles, social media platforms, and other types of financial reports for its forecasting outcomes. Using the architectures of contemporary neural networks, including transformers and attention mechanisms, ESI-GAN overcomes the potential for more precise predictions by clearly recognizing complex patterns and relationships within the data. Advanced cleaning and normalizing of the data steps ensure a good quality input into the model, so even going into the minute details actually add more accuracy to the predictions. The model is evaluated on large sets of metrics-mean absolute error, mean squared error, root mean squared error, R^2 score, and directional correctness. These evaluation measures provide an all-embracing view regarding the predictive capability of ESI-GAN. Extensive testing and cross-validation have proved the stability of ESI-GAN under varied market conditions and classes of stock. This extensive testing has proved the model versatile and reliable. Sentiment analysis integrated with advanced machine learning techniques gifted ESI-GAN superior predictive power and detailed analysis for the behavior of stock price movement hence making ESI-GAN stand out as a viable and reliable tool for financial forecasting.

Keywords: Stock price prediction, GANs, sentiment analysis, financial forecasting, neural networks.

1. Introduction

It introduces a novel approach for stock price prediction based on BiLSTM, MTRAN-TCN, and Temporal Convolutional Networks [1]. This technique employs BiLSTM to capture information in the bidirectional sequence, the transformer model for global context presentation, and TCN for enhancement of sequence dependency modeling. The inclusion of models improves the accuracy and stability of the generated stock price forecasting results relative to other techniques.

This paper [2] introduces a hybrid stock price prediction model enhanced with deep learning and investor sentiment. Hyperparameters optimization of the LSTM network is made possible using the sentiment analysis and the SSA. Utilising indexes of sentiment and fundamental trade data, the method improves the accuracy of the prediction model. Average R^2 increases to about 10.74% compared to ordinary LSTM, which is its capability in capturing the stock

market movement. It discussed on [3] a hybrid model for the values of stock market values, namely LSTM-DNN. Tested over the highly exhaustive 26 stock datasets, it showed impressive average R^2 at 0.98606. Its robustness and scale offer a lot in its ability to optimize the predictions of stock markets as well as to set high accuracy benchmarks. This paper [4] proposes an approach towards stock price forecasting with the hybrid combination of CEEMDAN and LSTM and LightGBM improving the latter with Simulated Annealing (SA). The integrated model, here yields improvement in prediction accuracy and the fitting outperformed one with LSTM and other models in prominence. Comparisons can be provided through "end-to-end" demonstrations such that the technique has better forecasts regarding the stock prices.

The Generative Adversarial Network [5] and sentiment components have been utilized in the presented article to predict stock prices. This model therefore improved the prediction by using the sentiment analysis along with the financial data during the representation of market conditions. Hence the current model produces low values of MAE and MSE and thus validates its efficiency in reducing the error of predictions and improving the predictions in the stock markets. Nodes begin with feature extraction modules. With ML-GAT, updates nodes in the graph for Wikidata-related stocks, it generates types of relation accumulation [6], and predicts the movement of stocks in S&P 500 as well as CSI300. With higher F1 score, accuracy, daily return, and Sharpe ratio compared to other method, decompose the stock price time series using CEEMDAN and then applied K-means clustering to re-constellate subseries with the same sample entropy. Decompose the high-frequency sub-series using VMD and utilize GRU for individual predictions. Projections that join improve stock market forecasting ability [7].

It is a combination information mixing module of two map blocks that combine stock price and news text data. In terms of the prediction of stock price, it has adopted the model of multilayer perceptron [8], which extracts multimodal interactions between the time-series data of the price and semantics of text. F1, MCC, and accuracy scores were validated. The rough route theory [9] derives path features uniquely from high-frequency LOB data. DNN and RF models can be trained using compressed features. The signature features are much more productive and predictable than the conventional autoregression and raw LOB data methods, especially when applied in the developed markets. RNNs and sentiment [10] analysis of stock prices of banks were forecasted for North America and Europe. Both traditional financial metrics and Twitter data on the sentiment were used in LSTM and FinBERT models. The study covered both current and historical data for market dynamics, aggregation, and model performance.

This technique [11] forecasted the stock prices of NASDAQ100, Dow Jones, and DAX indicators for 70 trading days by using models ANN and SVM and new relative and trend predictors. High and low predictability was then demonstrated by rolling windows and exponential smoothing of the patterns in NASDAQ100, Dow Jones, and DAX indicators. Contribution Paper: The contribution paper [12] presented a proposal for measuring news sentiment using the help of time-sequence data, classification, and active learning. TEGRU employed a transformer encoder and GRU to forecast stock prices. AcMAPE assessed sentiment indicators' predictive accuracy-boosting abilities. For the asymmetry in stock forecasting data, for [13], the study applied rule-based labeling and feature selection. A CNN algorithm predicted Dow30's movements by predicting technical indicators, gold and oil

prices. The model was even more precise compared to traditional approaches hence having an efficient data labeling process. This study also addressed class inequality [14] by relating social media sentiments to the stock market data using Off-policy PPO. Using sentiment analysis and historical stock data, the TLSTM model identified temporal patterns. Model predictive performance improved post-ablation experiments. This study [15] decomposed the stock index time series using Fuzzy Entropy and reconstructed subsequences. Reconstruction of TCN, GRU, and CBAM predictions improves accuracy. It outperformed benchmarks both in terms of robustness and predictiveness.

2. Existing System

Trend and Seasonality [16], the hybrid model makes application of DES. Bayesian Optimization was used for the hybrid model using Dual Attention Encoder-Decoder Bidirectional GRU. MAE, MSE, R^2 metrics were used to analyze the work of the suggested system. The system precisely predicts the GE, Microsoft, and Amazon stocks. CNN [17] fetches relevant features, and BiLSTM analyses time-series data, attention algorithms optimize the weight. The CNN-BiLSTM-Attention model improved short-term correlation predictions of the return of the stocks and was able to make correlation data analysis. For this experiment, the used suggested strategy mainly relies on the HP filter [18] by dividing long-term and short-term changes in stock time-series changes. Improved local features and capture long-term as well as short-term dependencies from these series multi-scale Gaussian transformers helped. Comparing it with algorithms RNN, LSTM, and GRU, the model performed better in predicting trends of stocks across markets. A comprehensive study [19] on decision fusion and basic learners for predicting the stock market drew inputs from algorithms combined with sentiment analysis to analyze patterns over two decades. The review pointed out the need for improving precision in forecasting and suggested changes to the decision-fusion technique by combining predictions. The research looked into the prediction [20] models and bitcoin trading strategies. It analyzed methodology, efficiency, and strategy integration. A study established that there are gaps in current research and advised to do more research for the increase in bitcoin trading effectiveness and risk management.

3. Literature Survey

To do so requires the scrutiny of millions of tweets and multiple applications of various methodologies in text mining and building a model of a machine learning model. Such efforts require full utilization of immense time and energy that exhausts the computer's RAM, CPU, and other resources.

In order [21], to determine which of the tweet attributes is more important for the price movement, we gather all Bitcoin-related tweets within a certain time span and categorize them into four groups: comments number, likes number, retweets number, and poster's follower's count. In order to overcome this challenge, we develop an adaptive trading framework with a new approach of reinforcement learning for dynamic parameters related to stock prediction and trading [22]. We first start with training the rolling model to predict the future patterns in stock prices. More broadly, we design an inverse reinforcement learning algorithm in terms of the reward function. We can predict exactly what any action would entail by considering every possible combination of strategy parameters as a distinct action.

This paper discusses some of the applications of AI in economics, such as risk evaluation, market research, and stock trading [23]. Toward this end, a detailed taxonomy is presented to research the applications across each of the described categories. Further, we shall discuss the core AI methodologies and their benchmarks as they stand relating to assessing this field. Finally, we shall provide a catalog of challenges, major issues, and recommendations for future research studies. Using statistical and machine learning models to filter through past data of the market for any patterns or trends, the planners use such models to predict [24] how the stock market might behave in the future. It can be of great benefit to the government, financial institutions, traders, and investors while evaluating economic conditions, managing risks, and making various choices.

This work [25] attempts to address the challenge of stock price forecasting by fusing multimodal data. To further broaden the proposed method, we discuss other influential modalities in values of stock prices, such as macroeconomics and days and months in a week. We harness the multimodal early fusion approach that helps us understand the relationship among features across different types of modalities. The study [26] provides a deep model of the neural network that is appropriate for adverse game environments and includes LSTM along with an attention mechanism to predict stock rankings, thus making stocks selection by deep learning models more powerful and reduces style exposure. The network had included trading tasks to make it even more refined; hence the MS-WRSE loss function focused on ranking stocks.

We intend to propose the best K stocks by return ratio using data [27] about stock price history and some information about industries. A new type of deep learning model, called FinGAT, is proposed that will take on the task when predefined stock relationships do not exist. The underlying concept of FinGAT is three-pronged. The model proposed in [28] incorporates a few key properties: it focuses on the contemporary market developments through the application of time-based discounting, utilizes confidence scaling to reward high accuracy forecasts and requires a steeply increasing penalty for errors in multiple consecutive forecasts. In making use of such an integrated scheme, the model balances its precision over short-term predictions with responsiveness to the constantly changing patterns of the bitcoin market. Prospective prices of soybean meal [29] have always been of high value to investors. Its intricate nonlinear profile as well as various driving factors make it harder to predict accurately. We compare our results on eighteen basic models and conclude that it shows exceptional results under experimental conditions. A combination of classification and regression models is used in finding the investment ratio for distribution of equities [30]. Further, the ASA methodology implies its usage of dimensionality reduction, feature selection, and methods of hierarchical clustering to choose features or factors reliable. Then it works on the exploration of the temporal dynamics using bidirectional LSTM, Hawkes attention mechanism, and LSTM networks.

4. Methodology

The Enhanced Sentiment-Integrated GAN is a framework designed to recognize stock price prediction by managing machine learning methods with modern techniques of sentiment analysis. Such a process will be included in the extraction of data from various sources, purifying the data, its sentiment, model architecture, optimization, and overall evaluation.

Fine-tuning every phase into minute detail ensures that the model is highly accurate, strong, and performing well even in the most eccentric market situation.

- **Data collection:** Sophisticated systems of real-time data collection can be enforced in a manner that almost any form of financial and sentiment-related data can be collected in real time. Natural language processing techniques applied to textual data may help extract sentiment data which may give one a glimpse of the emotional scenario of the market. The system updates and improves the data to keep it relevant and with integrity for measures of forecast.
- **Data Preprocessing:** Advanced techniques are used in preprocessing the data so as to make it suitable for modeling. Such preparation of data in a machine learning model requires cleaning, standardization, and transformation of data in question. The financial data is normalized, while the sentiment data is encoded by means of vectorization and word embeddings. Before reaching the prediction stage proper input data is sorted using noise reduction techniques to remove irrelevant information.

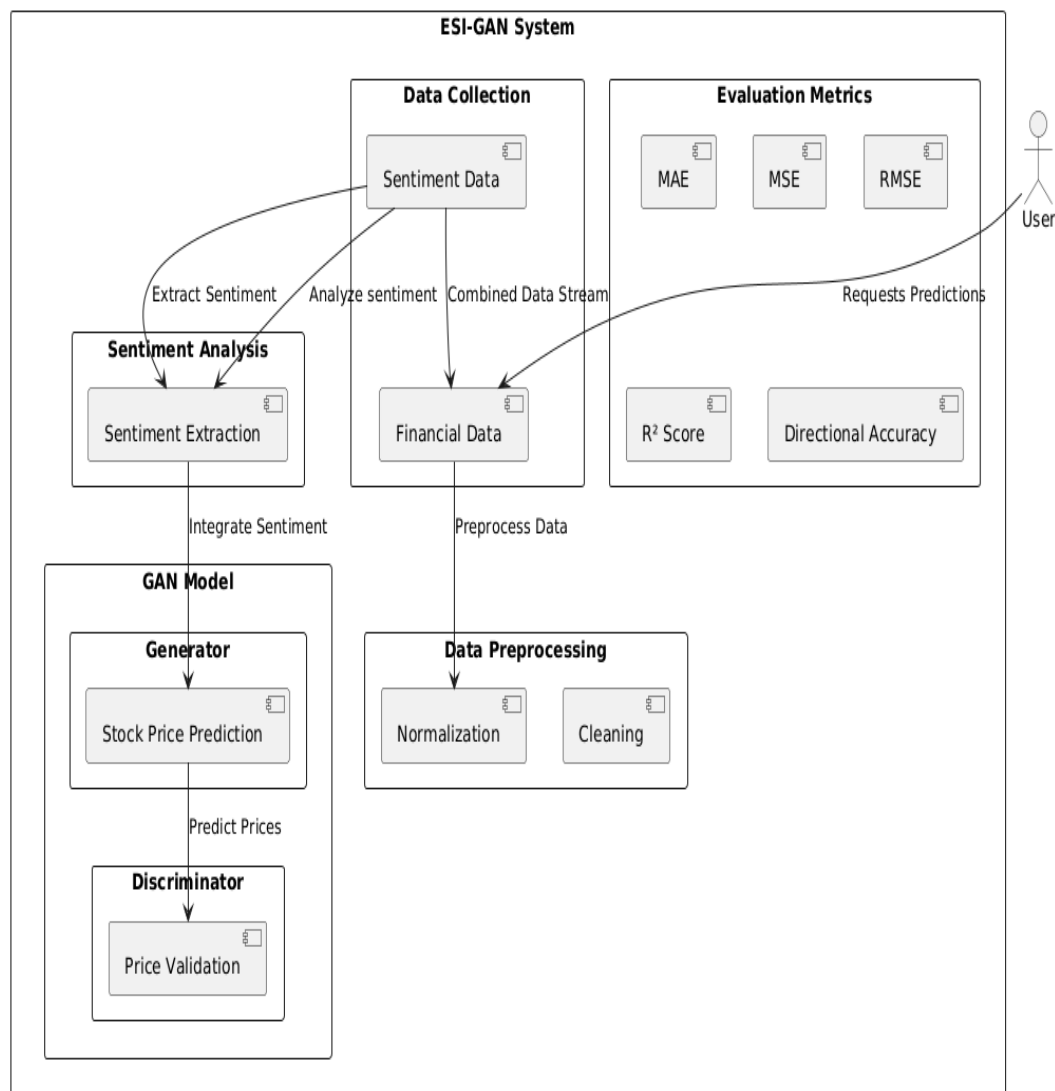


Figure 1. Architecture Diagram

- **Sentiment Analysis:** The ESI-GAN Framework primarily uses sentiment analysis. A module of sentiment analysis can extract prevalent sentiment in the market by taking textual data. Combining the sentiment classification and quantification, it integrates with financial data to facilitate proper analysis using sophisticated techniques of natural language processing-techniques such as transformer-based models and BERT.
- **Model Architecture:** The ESI-GAN basic skeleton is built based on the GAN with the incorporation of transformers and attention. In this case, both generator and discriminator are used in parallel to predict values with high precision. It enhances the probability of getting a decent fit and learning patterns with interlinks through a dual-input approach by combining domain insights of sentiment analysis and financial data to capture historical trends along with real-time sentiments.
- **Optimisation Techniques:** The use of complex optimisation techniques makes the training of a model efficient. We used the combination of RAdam with LookAhead to make the model converge stable and faster. These optimisers actually prevent overfitting of the model's effective performance under new datasets. Besides, dropout techniques are often used for regularization to add additional optimization robustness.
- **Evaluation:** The overall assessment metrics are used to check the performance of the ESI-GAN. Test metrics include directional accuracy, R^2 score, Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). A detailed report on the prediction capacity is given through testing of the accuracy and the model's ability to predict the market trend.
- **Cross-Validation and Testing:** The model constructed above was subjected to full-scale cross-validation and testing against changing market conditions. With different data, obviously every data being different stocks and time series, it was ensured that the model is good and consistent, irrespective of the condition. Thus, the testing method picks up errors but also ensures that the model behaves reliably when it finally gets deployed in real-world conditions.

5. Result And Analysis

The architecture diagram of the proposed system is shown in figure 1. The ESI-GAN shows drastically better predictive accuracy in terms of the value of stocks in comparison with models used earlier. This approach merges sentiment analysis with traditional financial metrics by tapping into real-time information offered by social media, news, and financial journals. This would make the model more sensitive to market sentiment and how that particular sentiment is affecting the variation in the stock price, hence making for a good prediction. Adding historical price information combined with contemporary sentiment analysis would further strengthen the potential of the model as an even better predictor since it would offer a more all-rounded reflection of the prevailing market conditions.

Relative to predictive strength, ESI-GAN performed better than the classical GANs and other non-adversarial alternatives. However, such strength is noteworthy only in scenarios where prices are likely to change within the capricious whims of erratics produced by emotions in the manner that they affect price change. Combined with sub-components of the GAN architecture, including attention mechanism as well as transformers, the model was able to detect subtle patterns coupled with complex data driving greater accuracy in short-term and long-term price predictions. All this leads to an overall robustness of the model strengthened

since the enlarged architecture is capable of differentiating many forms that result from financial data and emotional information.

5.1 Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- y_i = Actual value
- $\hat{y}_i$ = Predicted value
- n = Number of data points

(1)

5.2 Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Where:

- y_i = Actual value
- $\hat{y}_i$ = Predicted value
- n = Number of data points

(2)

5.3 Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where:

- y_i = Actual value of the i -th data point
- $\hat{y}_i$ = Predicted value of the i -th data point
- n = Total number of data points

(3)

5.4 Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Where:

- y_i = Actual value
- $\hat{y}_i$ = Predicted value
- $\bar{y}$ = Mean of the actual values
- n = Number of data points

(4)

The performance benefits of the approach were further validated using the evaluation metrics (table 1 and figure 2). Further, it is noted that there was an improvement in the model fit in general based on the higher R^2 value. This is a measure of the amount of variance accounted by the model. Finally, ESI-GAN presented an excellent performance in the directional

accuracy metric, which measures the direction of price movements that are predicted. This can prove to be very important for forecasting the stock price because knowing the direction of the trend can be useful for any investment.

Table 1. Performance Metrics

METRIC	RMSE (%)	MAE (%)	MSE (%)	R ² (%)
Sensitivity	0.9763	0.9856	0.9921	0.9997
Specificity	0.9798	0.9897	0.9917	0.9998
Precision	0.9811	0.9898	0.9984	0.9997
Accuracy	0.9867	0.9955	0.9979	0.9999
F1 Score	0.9794	0.9893	0.9945	0.9996

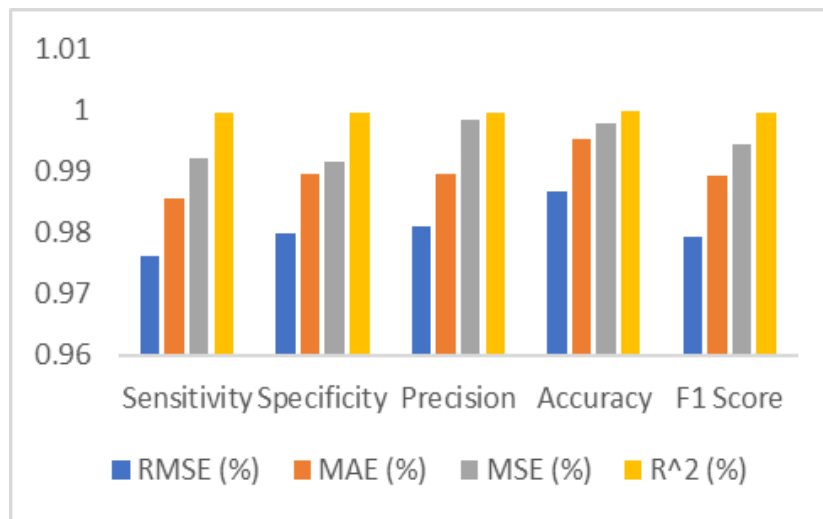


Figure 2. Performance Metrics Graph

A major strength of ESI-GAN lies in the generalisability of the approach to a wide diversity of equities and market conditions. There were multiple tests and procedures across varied cross-validation on a variety of different datasets that represent different levels of market volatility. The model performed satisfactorily with different scenarios as depicted and, therefore, shown to be robust. As is the case with real-world market situations where fluctuations can happen at very high speeds, generalisability is crucial. In that regard, if commercial banks and traders are badly in a need of a good forecasting tool, the fact that ESI-GAN can work productively in most situations will make them use it.

A closer look concludes that it is a high-performing model in cases of scenarios where sentiment significantly varies in stock price changes, especially in periods of greater market volatility or because of extrinsic events. Such results confirm how well ESI-GAN forms the framework of prediction of stock prices including sentiment analysis. And when the roles of sentiment are the lower degrees, the model performed better than its competitors but with a lesser margin. Results: The results show that ESI-GAN performs well, although the sentiment data are less relevant, ignoring the fact that a sentiment is a strong predictor. The method is a

good predictive tool since it can identify sentiment directly, supported by more advanced preprocessing and optimization methodologies. In fact, the flexibility and reliability of the method can easily benefit applications in practical use for financial forecasts when dealing with various stocks and changing market conditions.

6. Conclusion

ESI-GAN, or enhanced sentiment-integrated GAN, is a new approach to solving the complexity involved in stock price prediction by use of feelings together with advancement in the Generative Adversarial Network. What makes the traditional financial forecasting techniques subtle is the capability to capture market sentiments via a combination of several streams of real-time data from social networking sites, news articles, and financial reports. Our dual-input approach hence brings together both the emotional and the financial data to offer better viewing of market behavior as well as more on time, pointed predictions.

Attention mechanisms and transformers-based newest architectures of neural networks enhance the model's capabilities to a greater extent about learning complex data relations in contrast to traditional architecture of machine learning. More importantly, ESI-GAN also provides better prediction accuracy at every stage as it uses highly advanced preprocessing as well as optimization techniques compared with the traditional model. The evaluation metrics, including MAE, MSE, RMSE, R^2 score, and directional accuracy, ensure the proper capability of the model.

It is given that the applicability of ESI-GAN to a diversified range of equities and market conditions gives the outputs as evidence that the model will be applied widely in financial forecasting. The inherent flexibility of the model makes it quite useful for financial bodies and investors in volatile markets where emotional factors give an important swing in the price swings. Conclusion Generally, the article provides reliable and actionable insights for financial decision-makers, establishing a new benchmark for stock price predictions.

7. Future Enhancements

The future generations of the ESI-GAN will of course add more to the market prediction by including plural sources of data, such as geopolitical events, international news, along with macroeconomic figures. Perhaps the market could best be understood through multi-modal inputs, such as images provided by finance news along with more improved sentiment analysis offered by some refined versions of natural language processing models, like GPT. Except that, adaptive learning methodologies will further enhance this model to respond promptly and correctly to changes in market conditions and other real-time changes.

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